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UNIVERSITY'S SMART ID CARDS

NFC FLOWCHARTS

NFC SMART ID CARD SYSTEM

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- **Purpose:** Secure, smart identity verification for students and staff.
- **Workflow:** Card tap → 22-digit code → Server verification → Access granted/denied.
- **Frequency: Uses High Frequency (HF) 13.56 MHz NFC tech.**
- **Security:**
 - Encrypted NFC chips (e.g., MIFARE DESFire).
 - AES-256, SSH, HTTPS, PKI, and RBAC.
 - Admin alerts & logging.
- **System Structure:**
 - **Fronted:** Scanner + LCD.
 - **Middle ware:** Crypto + Validation.
 - **Backed:** UNI server + Library DB + Admin Dashboard.
- **Benefits:**
 - **Contactless & fast.**
 - **Real-time feedback.**
 - **Automated library system.**
 - **Salable & secure.**
- **Challenges:**
 - **High setup cost.**
 - **Complex integration.**
 - **Server dependency.**
 - **Crypto graphic management required.**

QUESTION 1

NFC ID CARD INFORMATION

KEYNOTE

In the modern era of digital transformation and smart campus innovations, managing identity securely and efficiently is a critical aspect of any educational institution. The university has introduced an advanced NFC-Based ID Card System designed to automate the identification of students and faculty, strengthen access control, and enhance library management processes. By leveraging High-Frequency (HF - 13.56 MHz) NFC technology, the system offers fast, contactless, and reliable identity verification, real-time data access, and secure processing, thereby creating an intelligent infrastructure for identity and attendance management.

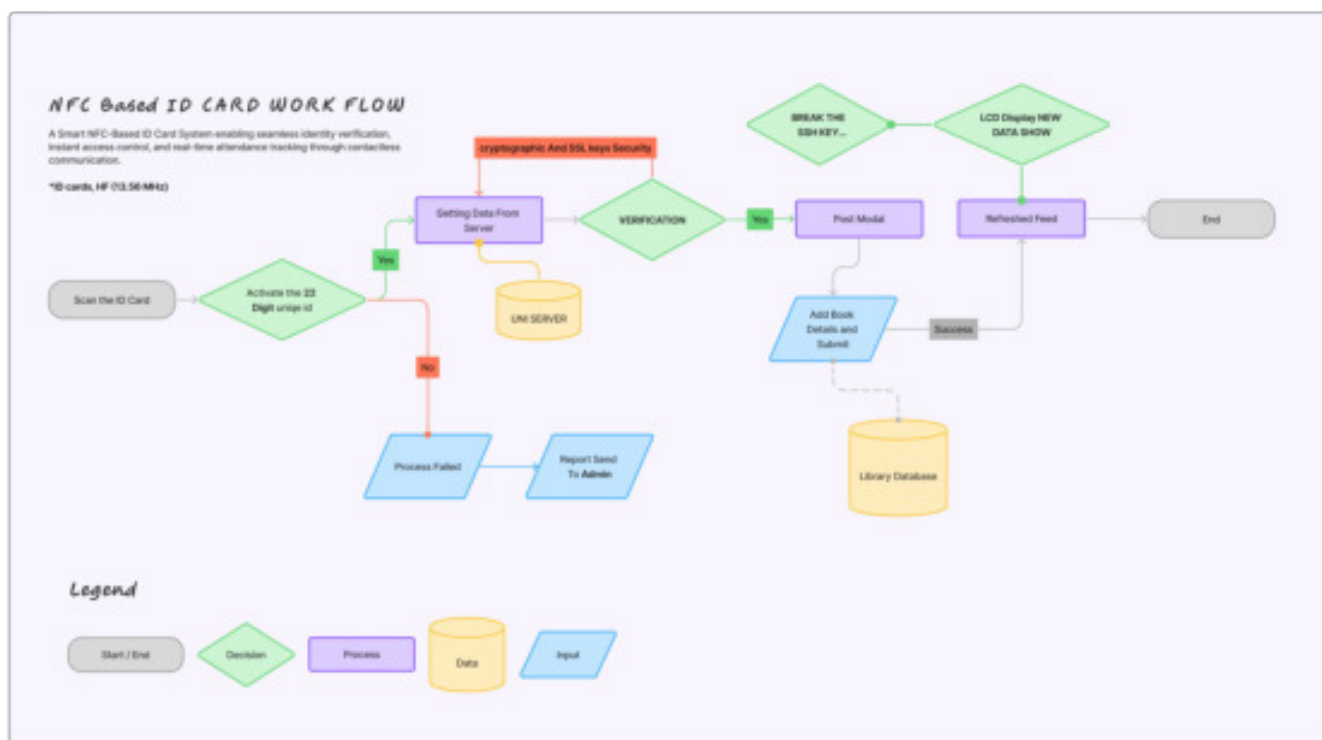


Fig 1.1 (Graph And Structure)

The entire workflow is engineered to ensure a smooth and secure experience, incorporating multiple layers of data verification and access control. It begins when a user scans their NFC ID card, initiating the activation of a unique 22-digit code embedded within the card. This code is then transmitted to the university's server for validation. The communication between the card reader and the server is encrypted using cryptographic techniques and SSH (Secure Shell) protocols to maintain data confidentiality and integrity. Once the identity is successfully verified, the user is granted access to additional functionalities, such as submitting book details to the library system. If the verification fails, the system logs the issue and automatically notifies the administrator for further investigation.

NFC ID CARD INFORMATION

This process integrates a frontend consisting of NFC-enabled scanners and LCD display units, a middleware layer featuring server-side validation, cryptographic security, and SSH authentication, and a backend that includes the university's central server (UNI SERVER), a library database (using platforms such as MySQL, PostgreSQL, or NoSQL), and an admin dashboard for real-time monitoring and reporting. The system ensures a well-defined flow of information: beginning with input via card tap, proceeding through server-side verification and cryptographic checks, and resulting in either successful access or an error report. The system also provides visual feedback on LCD screens and sends administrative alerts as needed.

Security is at the core of this system. Several layers of protection are implemented to safeguard against potential vulnerabilities. The use of encrypted NFC chips helps prevent cloning and unauthorized card duplication. Each card's unique 22-digit code is generated randomly to avoid predictability. Communication is conducted over secure HTTPS and SSH channels, while Public Key Infrastructure (PKI) and digital signatures are employed to validate the authenticity of data. Role-Based Access Control (RBAC) restricts access to critical features, and comprehensive logging mechanisms ensure all activities are monitored. Potential breakpoints include the risk of card cloning, the use of weak SSH keys, server-side data breaches, and unauthorized access to administrative interfaces. To mitigate these threats, it is recommended to use MIFARE DESFire EV2/EV3 cards, implement AES-256 encryption, and schedule regular security audits and key rotations.

The system presents numerous advantages. It supports contactless interactions, ensuring hygiene and user convenience. Verification processes are instantaneous, making the system efficient for daily use. Library operations are seamlessly integrated, and automated alerts ensure prompt administrative response. The system's cryptographic verification mechanisms make it highly secure, and the intuitive user interface, including real-time LCD feedback, enhances user experience. However, the system does pose certain challenges. The initial setup requires a significant financial investment in NFC hardware and secure cards. Integration across administrative and library systems can be complex. High server uptime is necessary to prevent service interruptions. Additionally, managing cryptographic keys and handling environmental factors such as signal interference require technical oversight. In conclusion, the NFC-based ID card system at the university represents a forward-thinking solution for secure identity management.